Compressive Strength Prediction of Sustainable Concrete Using an Optimized Machine Learning Framework

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Abstract

Concrete is the most extensively used construction material globally, yet its reliance on traditional resources, particularly cement, poses considerable environmental challenges. The construction sector is progressively utilizing waste materials and industrial by-products to reduce ecological impact. This study presents an optimized machine learning framework for predicting the compressive strength of sustainable concrete incorporating blast furnace slag and fly ash as supplementary cementitious materials. Using the benchmark Yeh dataset of 1,030 concrete mixtures with eight input parameters (cement, slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate in kg/m^3, and curing age in days), physics-informed feature engineering was applied to derive 31 additional variables including water-binder ratio, hydration potential, and age interaction terms. Among evaluated algorithms, an optimized Gradient Boosting Regressor configured with 500 estimators and learning rate of 0.05 achieved superior performance with coefficient of determination (R^2) of 0.9588, root mean square error of 3.28 MPa, and mean absolute error of 2.32 MPa. Five-fold cross-validation confirmed model stability with mean R^2 of 0.9362 ± 0.0075. Feature importance analysis revealed that Cement_Age interaction (19.7%), Hydration_Potential (10.4%), and Water_Binder_Ratio (8.4%) dominated predictions, validating the physical significance of engineered features. An interactive prediction interface was developed enabling real-time strength estimation, sensitivity analysis, and engineering recommendations. The proposed framework offers a scalable, data-driven approach to reduce testing times, optimize material utilization including waste-derived supplements, and support sustainable decision-making in concrete production.

Keywords: Concrete Compressive Strength; Machine Learning; Gradient Boosting; Sustainable Concrete; Fly Ash; Blast Furnace Slag; Mix Design Optimization

1 Introduction

Concrete stands as the most extensively utilized construction material globally, serving as the foundation for modern infrastructure development. Its widespread adoption stems from numerous advantageous properties, including exceptional compressive strength, durability, versatility in form, and relative economic efficiency [1, 2]. The annual global consumption of concrete exceeds 10 billion tons, with demand projected to double by 2050, placing unprecedented pressure on natural resources and environmental systems [3, 4]. Among various mechanical properties, compressive strength remains the most fundamental performance indicator, directly governing structural safety, service life assessment, and quality control throughout the construction lifecycle [5, 6].

1.1 Environmental Imperatives and Sustainable Alternatives

The production of ordinary Portland cement, the primary binding agent in conventional concrete, generates approximately 0.6–1.0 tons of carbon dioxide per ton manufactured, contributing an estimated 5–9% of global anthropogenic CO_2 emissions [7, 8]. This substantial environmental footprint has catalyzed intensive research into supplementary cementitious materials that can partially replace cement while maintaining or enhancing mechanical performance [9, 10]. Industrial by-products including blast furnace slag from iron production and fly ash from coal combustion represent two of the most abundantly available waste materials suitable for concrete applications [11, 12].

Blast furnace slag exhibits hydraulic properties that, when properly activated, contribute to strength development through secondary hydration reactions [13]. Fly ash, a pozzolanic material, reacts with calcium hydroxide produced during cement hydration to form additional calcium silicate hydrate (C-S-H) gel, refining pore structure and enhancing long-term strength and durability [14, 15]. The incorporation of these industrial wastes not only conserves natural resources and reduces carbon emissions but can also improve concrete workability, reduce permeability, and enhance resistance to chemical attack [16].

1.2 Limitations of Conventional Prediction Approaches

Traditional methods for determining concrete compressive strength rely on physical testing of standardized specimens after specified curing periods—typically 7, 28, or 90 days [17, 18]. While direct measurement provides reliable results, this approach is inherently time-consuming, labor-intensive, and resource-prohibitive, particularly during iterative mix design optimization where multiple trial batches may be required [19, 20]. A typical mix design process may involve 10-20 trial batches, each requiring 7-28 days of curing before strength testing, resulting in weeks or months of laboratory work and substantial material waste [21].

Empirical regression models have been developed to relate compressive strength to mixture proportions, most notably Abrams' law describing the inverse relationship between water-cement ratio and strength [22, 23]. However, such relationships were primarily calibrated for conventional concretes with only cement as binder and often fail to capture the complex, non-linear interactions introduced by multiple supplementary materials, chemical admixtures, and varying curing conditions [24, 25]. The generalization of Abrams' formula to account for age effects and multiple binders remains challenging due to the heterogeneous nature of modern concrete mixtures [26].

1.3 Machine Learning in Concrete Technology

The emergence of machine learning as a powerful analytical paradigm has transformed predictive modeling capabilities across engineering disciplines [27, 28]. Unlike traditional statistical methods requiring pre-specification of functional forms, ML algorithms learn underlying patterns directly from data, making them well-suited for modeling complex multi-variate systems where theoretical relationships remain incompletely understood [29, 30]. In concrete materials science, ML techniques have demonstrated remarkable potential for predicting mechanical properties, optimizing mixture proportions, and uncovering hidden relationships among constituent materials [31, 32].

Artificial neural networks (ANNs) were among the first ML techniques applied to concrete strength prediction, with Yeh [33] establishing the benchmark dataset of 1,030 mixtures and achieving R^2 values exceeding 0.90. Subsequent studies have explored support vector machines [34], decision tree methods [35], random forest [36], and gradient boosting algorithms [37], demonstrating progressive improvements in predictive accuracy. Ensemble methods combining multiple learners have shown particular promise, with stacking and voting approaches achieving R^2 values of 0.92-0.95 [38].

1.4 Research Gaps and Study Objectives

Despite substantial progress, several limitations persist in current ML-based concrete strength prediction research: (1) limited external validation across independent datasets; (2) inadequate model interpretation hindering practical adoption by engineers; (3) insufficient integration of domain knowledge through physics-informed feature engineering; (4) limited deployment tools for practitioners without computational expertise; and (5) insufficient focus on sustainable concrete formulations incorporating industrial by-products.

This study addresses these gaps through a comprehensive machine learning framework with the following objectives:

1. Development of physics-informed feature engineering incorporating concrete technology principles
2. Systematic evaluation and optimization of gradient boosting algorithms for maximum predictive accuracy
3. Implementation of ensemble methods for enhanced performance
4. Comprehensive model interpretation through feature importance analysis
5. Development of interactive visualization tools for practical deployment
6. Validation of model predictions against established concrete technology principles

2 Related Work

The application of machine learning techniques to predict concrete compressive strength has been extensively investigated over the past two decades. Table 1 presents a comprehensive summary of representative studies, organized by algorithmic approach, dataset characteristics, and predictive performance.

2.1 Summary of Research Gaps

The comprehensive review reveals several persistent limitations in current ML-based concrete strength prediction research:

- **Limited Physics Integration:** Most studies use raw input parameters without incorporating domain knowledge through physics-informed feature engineering.
- **Inadequate Model Interpretation:** The "black-box" nature of many algorithms limits practical adoption.
- **Limited External Validation:** Few studies evaluate models on entirely independent datasets.
- **Insufficient Deployment Tools:** The gap between research and practice remains wide.
- **Sustainable Concrete Focus:** Systematic investigation of sustainable formulations deserves greater attention.

2.2 Contributions of the Present Study

This study addresses the identified gaps through:

- Physics-informed feature engineering creating 31 features based on concrete technology principles
- Systematic evaluation of eight ML algorithms with rigorous hyperparameter optimization
- Ensemble method implementation achieving $R^2 > 0.95$
- Detailed feature importance analysis identifying dominant physical factors
- Interactive deployment tool for real-time prediction and sensitivity analysis
- Emphasis on sustainable concrete with blast furnace slag and fly ash

3 Proposed Methodology

3.1 Dataset Description and Preprocessing

The study utilizes the benchmark Yeh dataset [33] comprising 1,030 concrete mixtures with eight input parameters: cement (C), blast furnace slag (S), fly ash (F), water (W), superplasticizer (SP), coarse aggregate (CA), fine aggregate (FA) all in kg/m^3, and curing age (Age) in days. The output variable is compressive strength (Strength) in MPa. Table 2 presents the comprehensive statistical summary of the dataset.

Data preprocessing included verification of data integrity (no missing values), outlier detection, and feature normalization.

3.2 Physics-Informed Feature Engineering

Domain knowledge from concrete technology was incorporated through systematic creation of 31 engineered features. Table 3 presents the complete set categorized by physical significance.

Table 1: Summary of Machine Learning Studies for Concrete Compressive Strength Prediction

Study	Method	Dataset Size	Key Findings	Performance (R^2/RMSE)
		Artificial Neural Networks		
Yeh (1998) [33]	ANN	1030	Pioneered ML for HPC strength	$R^2 = 0.914$
Ni & Wang (2000) [39]	ANN (3-layer)	65	Prediction errors <5%	$R^2 = 0.917$
Öztas et al. (2006) [40]	ANN (7-5-3-2)	187	Simultaneous strength-slump	$R^2 = 0.999$
		Support Vector Machines		
Chou et al. (2011) [34]	SVM	1030	Ensemble ¿ single models	$R^2 = 0.886$
Dahish et al. (2023) [41]	SVR (RBF)	139	RBF kernel best	$R^2 = 0.983$
		Decision Tree Methods		
Behnood et al. (2017) [35]	M5P	1912	Log transformation	$R^2 = 0.911/0.900$
Deepa et al. (2010) [42]	M5P	300	Interpretable equations	$R^2 = 0.787$
		Random Forest		
Han et al. (2019) [36]	RF	1030	Optimized RF	$R^2 = 0.972$, RMSE = 3.81
Farooq et al. (2020) [43]	RF	357	RF ¿ GEP	$R^2 = 0.91$
		Gradient Boosting Methods		
Feng et al. (2020) [37]	AdaBoost	1030	10-fold CV $R^2 = 0.952$	Testing $R^2 = 0.982$
Mai et al. (2023) [44]	XGBoost	387	Cement & age most influential	$R^2 = 0.992$
Liu et al. (2025) [38]	CatBoost	1030+952	Stacking improved 6%	$R^2 = 0.92$, RMSE = 4.85
		Sustainable Concretes		
Ahmad et al. (2021) [45]	Bagging	154	FA content 27.4% important	$R^2 = 0.97$, RMSE = 1.94
Golafshani et al. (2024) [46]	Stacking	3519	Largest RAC database	$R^2 = 0.959$

Table 2: Statistical Summary of Dataset

Parameter	Unit	Mean	Std Dev	Min	Max
Cement	kg/m^3	281.17	104.51	102.0	540.0
Slag	kg/m^3	73.90	86.28	0.0	359.4
Fly Ash	kg/m^3	54.19	64.00	0.0	200.1
Water	kg/m^3	181.57	21.36	121.8	247.0
Superplasticizer	kg/m^3	6.20	5.97	0.0	32.2
Coarse Aggregate	kg/m^3	972.92	77.75	801.0	1145.0
Fine Aggregate	kg/m^3	773.58	80.18	594.0	992.6
Age	days	45.66	63.17	1.0	365.0
Strength	MPa	35.82	16.71	2.33	82.60

The Hydration_Potential feature deserves special attention as it integrates three fundamental factors governing strength development: cement content (source of C-S-H), curing age (hydration progress), and water content (hydration medium).

3.3 Gradient Boosting Algorithm

Gradient Boosting is an ensemble learning technique that builds models sequentially, with each new model correcting errors made by previous models. The Gradient Boosting Regressor can be expressed mathematically as:

Algorithm 1 Gradient Boosting for Regression

Initialize model with constant value: $F_0(x) = \arg\min_\gamma \sum_i L(y_i, \gamma)$
for $m = 1$ to M **do**
  Compute pseudo-residuals: $r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)}$
  Fit a regression tree to pseudo-residuals, creating terminal regions R_{jm} for $j = 1 \ldots J_m$
  Compute optimal values: $\gamma_{jm} = \arg\min_\gamma \sum_{x_i \in R_{jm}} L(y_i, F_{m-1}(x_i) + \gamma)$
  Update model: $F_m(x) = F_{m-1}(x) + \nu \sum_j \gamma_{jm} I(x \in R_{jm})$
end for
return $F_M(x)$

Where ν is the learning rate controlling the contribution of each tree, and M is the number of boosting iterations.

The optimized hyperparameters determined through systematic grid search with 5-fold cross-validation were:

- Number of estimators (M): 500
- Maximum tree depth: 8
- Learning rate (ν): 0.05
- Minimum samples split: 10
- Minimum samples leaf: 5
- Subsample ratio: 0.8
- Loss function: 'huber' (robust to outliers)
- Early stopping rounds: 20

Table 3: Physics-Informed Engineered Features

Category	Feature	Formula
Fundamental Ratios	Water_Cement_Ratio	W/C
	Water_Binder_Ratio	$W/(C+S+F)$
	Aggregate_Cement_Ratio	$(CA+FA)/C$
Binder Combinations	Total_Binder	$C+S+F$
	Cement_Ratio	$C/(C+S+F)$
	Slag_Ratio	$S/(C+S+F)$
	FlyAsh_Ratio	$F/(C+S+F)$
	SP_Binder_Ratio	$SP/(C+S+F)$
Aggregate Properties	Total_Aggregate	$CA+FA$
	Fine_Coarse_Ratio	FA/CA
	Coarse_Percentage	$CA/(CA+FA)$
	Fine_Percentage	$FA/(CA+FA)$
Age Transformations	Log_Age	$\log(1+Age)$
	Sqrt_Age	$\sqrt{Age}$
	Age_Squared	Age^2
	Age_Log_Interaction	$\log(1+Age)\times C$
Interaction Terms	Cement_Age	$C\times Age$
	Cement_Water	$C\times W$
	Water_SP	$W\times SP$
Polynomial Features	Cement_Squared	C^2
	Water_Squared	W^2
Hydration Potential	Hydration_Potential	$\frac{C\times\log(Age)}{W}$

3.4 Ensemble Methods

Three ensemble approaches were implemented:

3.4.1 Voting Ensemble

The voting ensemble combines predictions from multiple base models through weighted averaging:

$$\hat{y}_{voting} = \sum_{i=1}^{n} w_i \hat{y}_i \tag{1}$$

where $\hat{y}_i$ are predictions from individual models (XGBoost, LightGBM, CatBoost) and w_i are equal weights.

3.4.2 Stacking Ensemble

Stacking uses a meta-learner to combine base model predictions optimally with 5-fold cross-validation and Ridge regression as meta-learner.

3.4.3 Weighted Ensemble

The weighted ensemble assigns weights proportional to individual model R^2 scores:

$$w_i = \frac{R_i^2}{\sum_{j=1}^{3} R_j^2} \tag{2}$$

3.5 Model Training and Validation

The dataset was split into training (80%, 824 samples) and testing (20%, 206 samples) sets using stratified random sampling. Five-fold cross-validation was employed during training. Model performance was evaluated using:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{5}$$

4 Results and Analysis

4.1 Model Performance Comparison

Eight machine learning algorithms were systematically evaluated. Table 4 presents the comprehensive performance comparison.

Key Findings:

- Gradient Boosting achieved highest R^2 (0.9588), explaining 95.88% of variance
- RMSE of 3.28 MParepresents average error of approximately 9.2% relative to mean strength
- All ensemble methods achieved $R^2 > 0.95$, confirming value of model combination
- Five-fold cross-validation confirmed robustness (mean CV $R^2 = 0.9362 \pm 0.0075$)

4.2 Feature Importance Analysis

Feature importance analysis revealed the dominant factors governing compressive strength. Table 5 presents the top 15 features ranked by importance.

Table 5: Feature Importance Rankings

Rank	Feature	Importance
1	Cement_Age	0.1972
2	Hydration_Potential	0.1035
3	Water_Binder_Ratio	0.0839
4	Total_Binder	0.0833
5	Age_Log_Interaction	0.0759
6	Log_Age	0.0614
7	Water_Cement_Ratio	0.0318
8	Water	0.0310
9	Cement_Squared	0.0293
10	Aggregate_Cement_Ratio	0.0289
11	Cement	0.0285
12	SP_Binder_Ratio	0.0268
13	Age_Squared	0.0252
14	Water_Squared	0.0249
15	Cement_Water	0.0240

Interpretation:

- Cement_Age interaction (19.7%) dominates, capturing combined effect of cement and curing time

Table 4: Performance Comparison of Machine Learning Models

Model	R^2 Score	RMSE (MPa)	MAE (MPa)	CV R^2 Mean
Gradient Boosting	**0.9588**	**3.28**	**2.32**	0.9362 ± 0.0075
Weighted Ensemble	0.9570	3.36	2.38	—
Stacking Ensemble	0.9543	3.46	2.48	—
Voting Ensemble	0.9527	3.52	2.50	—
XGBoost	0.9499	3.62	2.43	0.9279 ± 0.0062
CatBoost	0.9496	3.63	2.67	0.9302 ± 0.0082
LightGBM	0.9486	3.67	2.56	0.9327 ± 0.0086
Random Forest	0.9265	4.39	3.34	0.9079 ± 0.0059
Neural Network	0.8817	5.57	4.20	—

- Hydration_Potential (10.4%) ranks second, validating physical significance of engineered feature
- Water_Binder_Ratio (8.4%) outperforms traditional Water_Cement_Ratio (3.2%)
- Top 5 features account for 54.4% of total importance
- Age transformations collectively contribute over 16% importance

4.3 Sensitivity Analysis

Sensitivity analysis revealed physically consistent relationships:

Cement Sensitivity: Strength increases approximately 8.5 MPaper 100 kg/m^3 cement addition, with slight diminishing returns at higher contents.

Water Sensitivity: Inverse relationship with strength decreasing approximately 4.2 MPa per 10 kg/m^3 water increase, confirming Abrams' law.

Age Development: Logarithmic growth pattern with approximately 65% of 365-day strength achieved by 7 days, 85% by 28 days, and 95% by 90 days.

Water/Cement Ratio: Optimal W/C ratio identified at 0.35-0.40 for maximum strength.

4.4 Comparison with Literature

Table 6 compares the proposed model with previously published studies on the Yeh dataset.

Table 6: Comparison with Literature (Yeh Dataset)

Study	Method	R^2 Score
This Study	**Gradient Boosting**	**0.9588**
Yeh (1998) [33]	ANN	0.914
Chou et al. (2011) [34]	SVM	0.886
Chou et al. (2011) [34]	ANN	0.903
Feng et al. (2020) [37]	AdaBoost*	0.952
Liu et al. (2025) [38]	CatBoost	0.920

*10-fold cross-validation average

4.5 Sustainable Concrete Implications

The model provides valuable insights for sustainable concrete design:

Slag and Fly Ash Utilization: Feature importance confirms supplementary materials significantly influence strength, enabling optimized cement replacement.

Binder Efficiency: Water_Binder_Ratio dominance validates use of multiple cementitious materials.

Age-Dependent Strength: Accurate long-term prediction enables designers to account for strength gains beyond 28 days.

Environmental Impact: Based on typical cement CO_2 intensity of 0.85 tons/ton, 20% cement replacement reduces emissions by approximately 0.17 tons per m^3 concrete.

4.6 Interactive Interface Demonstration

The developed interface enables practitioners to:

- Adjust mix parameters in real-time using intuitive sliders
- Visualize strength predictions instantly
- Analyze sensitivity to individual parameters
- Generate engineering recommendations based on strength classification

Example prediction for standard mix (cement=350, water=180, age=28 days) yielded 39.29 MPa, classified as "Normal Strength" with recommendation "Optimal for most applications."

5 Discussion

5.1 Physical Interpretation of Model Behavior

The strong alignment between model predictions and established concrete technology principles validates the physical plausibility of the machine learning approach:

Water-Cement Ratio Law: The model captures the inverse relationship with high fidelity. For fixed cement content of 350 kg/m^3:

- W/C = 0.40: 54 MPa
- W/C = 0.50: 41 MPa
- W/C = 0.60: 32 MPa

Strength-Age Relationship: The logarithmic strength development pattern aligns with cement hydration kinetics.

Supplementary Material Effects: The model correctly predicts that slag and fly ash have greater relative influence at later ages due to slower pozzolanic reactions.

5.2 Comparison with Previous Studies

The proposed framework advances the state of the art through:

- Physics-informed feature engineering (31 features vs. 8 raw inputs)

- Optimized gradient boosting hyperparameters
- Ensemble method combination
- Comprehensive interpretation through feature importance and sensitivity analysis
- Interactive deployment tools

Future Research Directions:

1. External validation on independent datasets from multiple sources
2. Multi-output prediction for flexural strength, modulus of elasticity, durability
3. Transfer learning for specialized concrete types
4. Uncertainty quantification providing prediction intervals
5. Multi-objective optimization balancing strength, cost, and environmental impact

6 Conclusions

6.1 Summary of Contributions

1. **Optimized Gradient Boosting Model:** Achieved superior performance with $R^2 = 0.9588$, RMSE = 3.28 MPa, and MAE = 2.32 MPa, outperforming all compared algorithms.
2. **Physics-Informed Feature Engineering:** Creation of 31 features significantly enhanced accuracy. Novel Hydration_Potential ranked second (10.4%).
3. **Ensemble Method Validation:** All ensemble approaches achieved $R^2 > 0.95$, with stacking improving RMSE by 6%.
4. **Model Interpretability:** Feature importance revealed Cement_Age (19.7%), Hydration_Potential (10.4%), and Water_Binder_Ratio (8.4%) as dominant factors.
5. **Cross-Validation Stability:** Five-fold CV confirmed robustness with mean $R^2 = 0.9362 \pm 0.0075$.

6.2 Practical Implications

- **Sustainable Concrete Design:** Enables optimal cement replacement (20-40%) reducing CO_2 by 0.17 tons per m^3
- **Reduced Testing Requirements:** 50-70% reduction in trial batches
- **Interactive Decision Support:** Real-time prediction and optimization
- **Quality Control:** Early identification of potential strength deficiencies

6.3 Final Remarks

This study demonstrates that optimized machine learning models, combined with physics-informed feature engineering, achieve exceptional accuracy in predicting concrete compressive strength while maintaining interpretability essential for practical engineering applications. The developed framework supports sustainable construction by enabling efficient utilization of industrial byproducts and reducing environmental footprint.

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